FEATURE ENGINEERING

* ML models only understand numbers

$f(x) = y$

mathematical

human

ML

x → features

f

y → label

numerical ??

machine learning

$$X = \begin{bmatrix} x_1 & , & x_2 & , & x_3 & , & x_4 \\ km & & \text{model} & & \text{year} & & \text{fuel} \\ \checkmark & & \text{categorical} & & \checkmark & & \begin{bmatrix} \text{petrol} \\ \text{diesel} \\ \text{electric} \end{bmatrix} \end{bmatrix}$$

Categorical encoding: to convert cat-features into numeric features.

Label encoding

fuel		P	D	E	H	null	weight
P →	0	1	0	0	0		23
D →	1	0	1	0	0		40
E →	2	0	0	1	0		50
H →	3	0	0	0	1		60
null →							

(one-hot encoding
dummy variable)

1000
0100
0010
0001

$$\begin{bmatrix} A \\ B \\ C \\ D \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

label encoding

VARIABLE

$$\begin{bmatrix} D \\ E \\ F \end{bmatrix} \qquad \begin{bmatrix} 5 \\ 6 \end{bmatrix} \checkmark$$

ORDINAL

ordered categories

$A > B > C > \dots > F$

price

$$\begin{bmatrix} [\ ] \\ [\ ] \\ [\ ] \end{bmatrix}$$

S → 0
M → 1
L → 2

car - price

	km	old [0-10]
	20,000	1

200,

50,000

80,000

3

4

Scaling

not to use on outlier data

Standard scaling

mean = m

S.D :

100,00

20,000

mean

Min Max Scaler : Normalization

X_1		X_2	
4000	0	30	0
⋮	⋮	⋮	⋮
40,000	1	80	1
	[0-1]		[0-1]

$$\frac{x - \min}{\max - \min}$$

Robust Scaler

Imbalance : classification

↓

fraud detection

↓

credit card

99 % transaction → fraud

class imbalance

non-fraud

1% → fraud.

under sampling.

train data

70	30

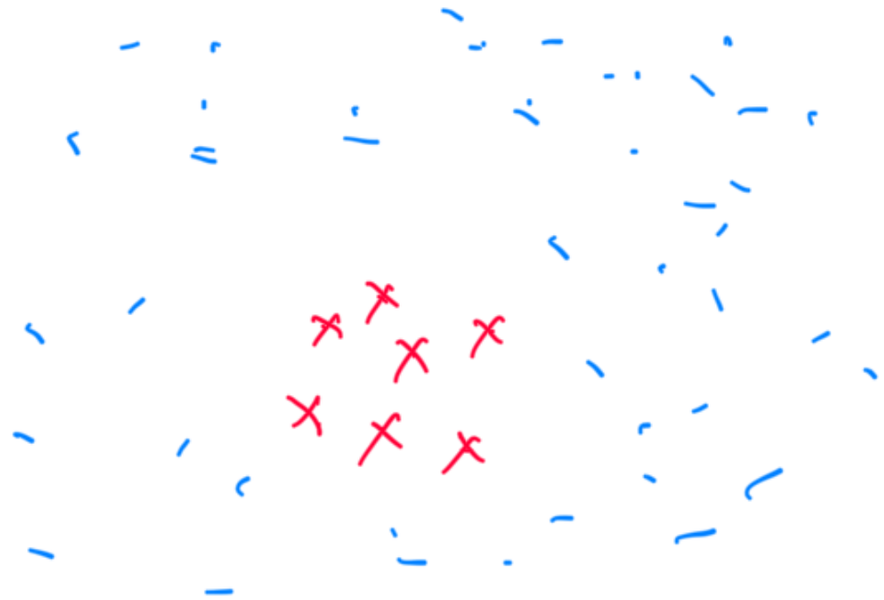
10,000	N.F

500	F

2000	500

SMOTE :

Synthetic minority oversampling technique.



Logistic Regression classification.

Data leakage :

training ✓ accuracy ↑

BMI

exerc	food	age	weight

testing ∝ accuracy ↓

	user label	fraud
	1	1
	0	0
	0	0
	0	0
	1	1

$\boxed{99\%}$

user label

0 → 0

House price prediction

X

y

$[x_1, x_2, x_3 \ldots]$

$\boxed{\text{M.L}}$:

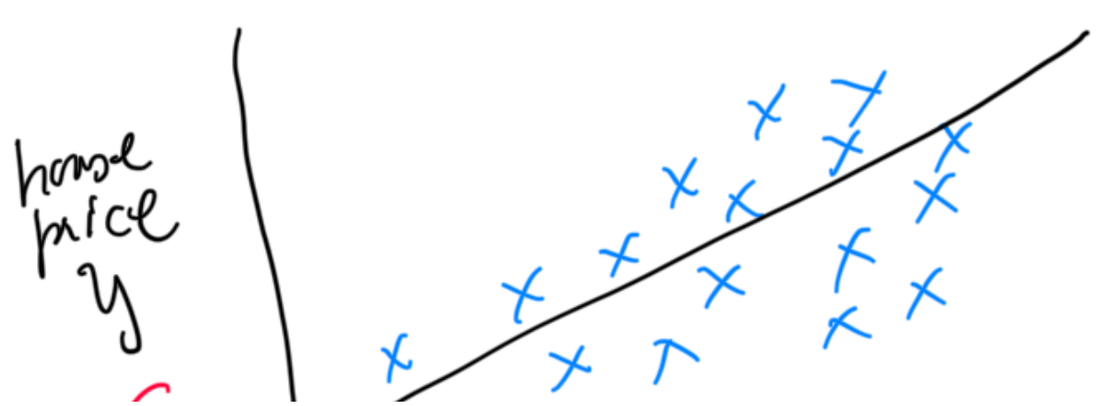
$$\boxed{y = f(x)}$$

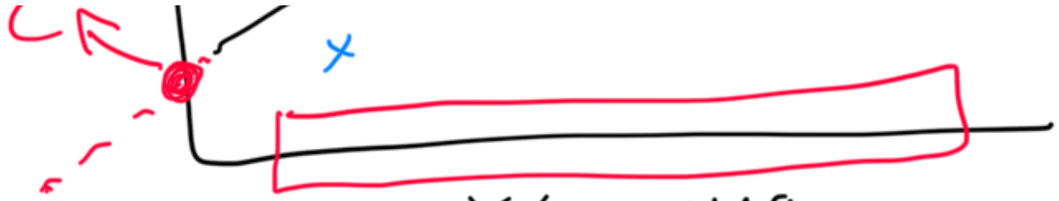
$x \rightarrow \dfrac{\text{area}}{\text{square feet}}$

Linear relationship

straight line

(Linear - Regression)

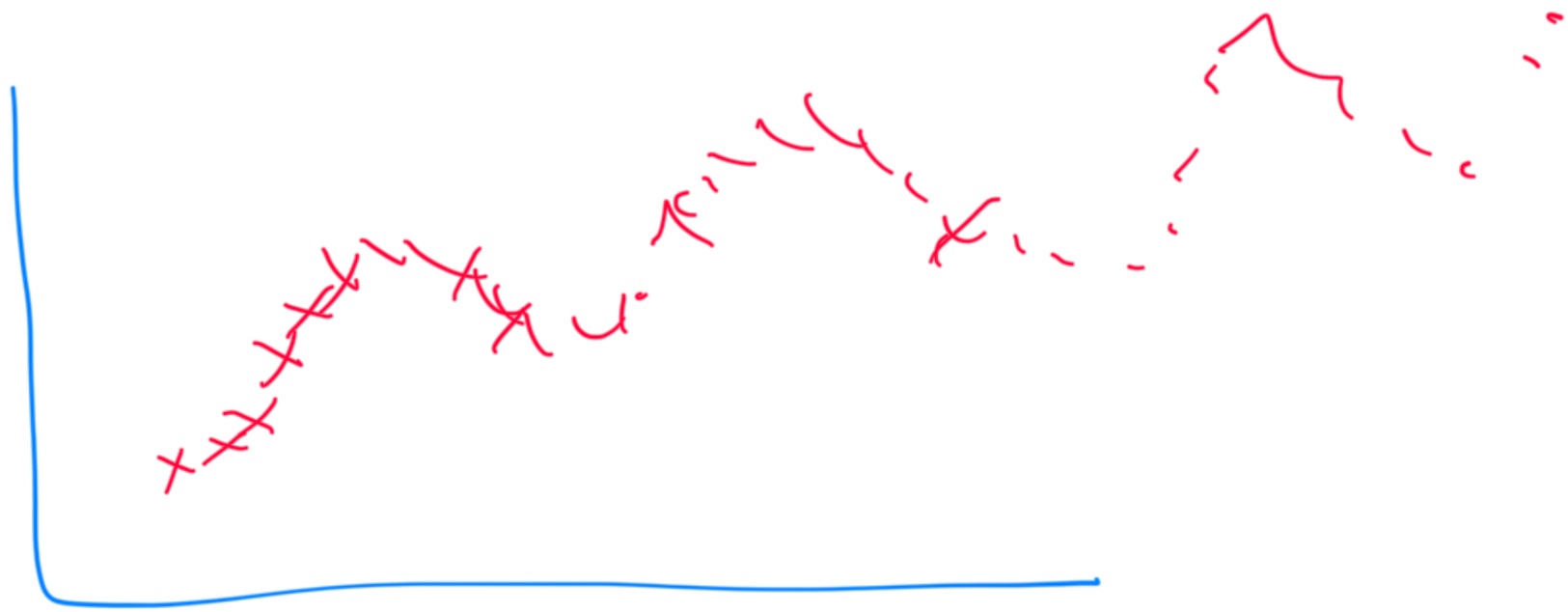




X : area

$$\boxed{y = mx + c}$$

best fit line



X : 1 variable → simple linear regression

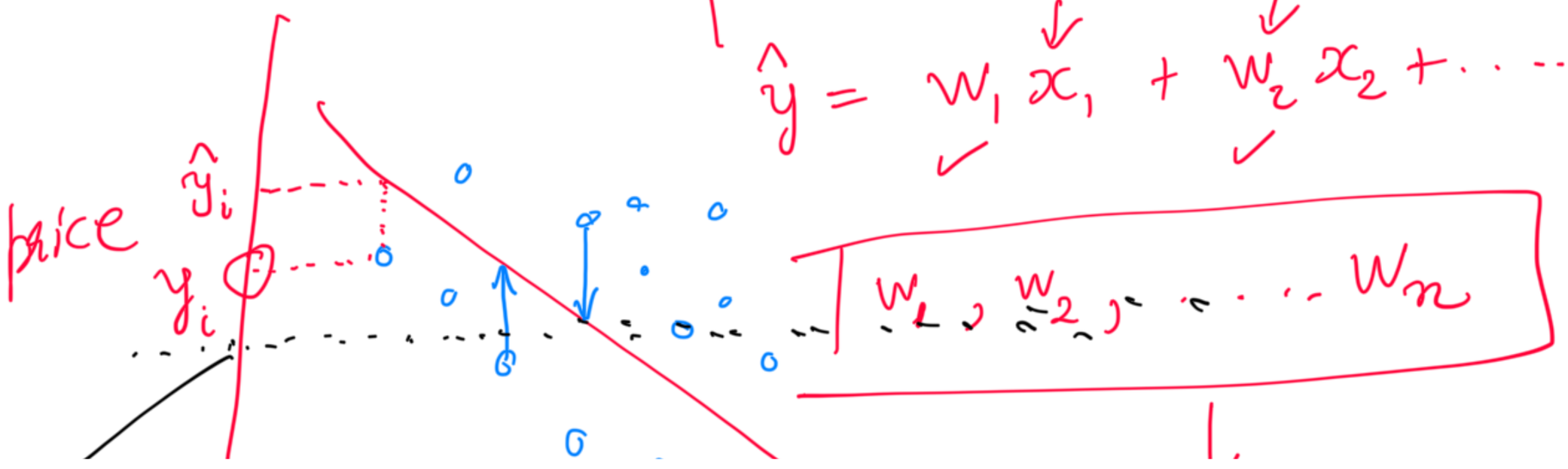
X : 2+ → multiple linear

regression.

$$y = f(x)$$

$$y = m\boxed{x} + c$$

$$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots + \boxed{w_0}$$



this is what ML is learning.

MSE.

$\hat{y} \rightarrow$ predicted

$y \rightarrow$ actual.

total error : $\sum (y - \hat{y}) \approx \boxed{0}$

$$\left[\frac{1}{n} \sum (y - \hat{y})^2\right] \quad \text{SSE}$$

MSE : mean squared error

↳ aim of LR is to minimize.

$$\hat{sales} = \boxed{w_1} \times radio + \boxed{w_2} \times TV + \boxed{w_3} \times newspaper + \boxed{w_0}$$

coefficients

$$\text{sales} = \underline{5} \times \text{radio} + \underline{3} \times TV + \underline{0.5} \times NP + \underline{7}$$

coefficient → represent feature importance.

$$\boxed{\hat{y} = 7x_1 + 2x_2 - 9\underbrace{x_3} + 1.5}$$

$$\boxed{0}$$

Model — car price prediction

↓

only — final prediction — matters

[feature importance
model interpretation]

model 2

model 1

TSE

91

MSE

150

comp 1

comp 2

profit

20 cr

100 cr

base investment

20 cr

1000 cr.

100 %

10 %

$$\frac{\text{profit}}{C.P} = \frac{SP - C.P}{C.P}$$

MSE

$\rightarrow$ 5

best fit ✓

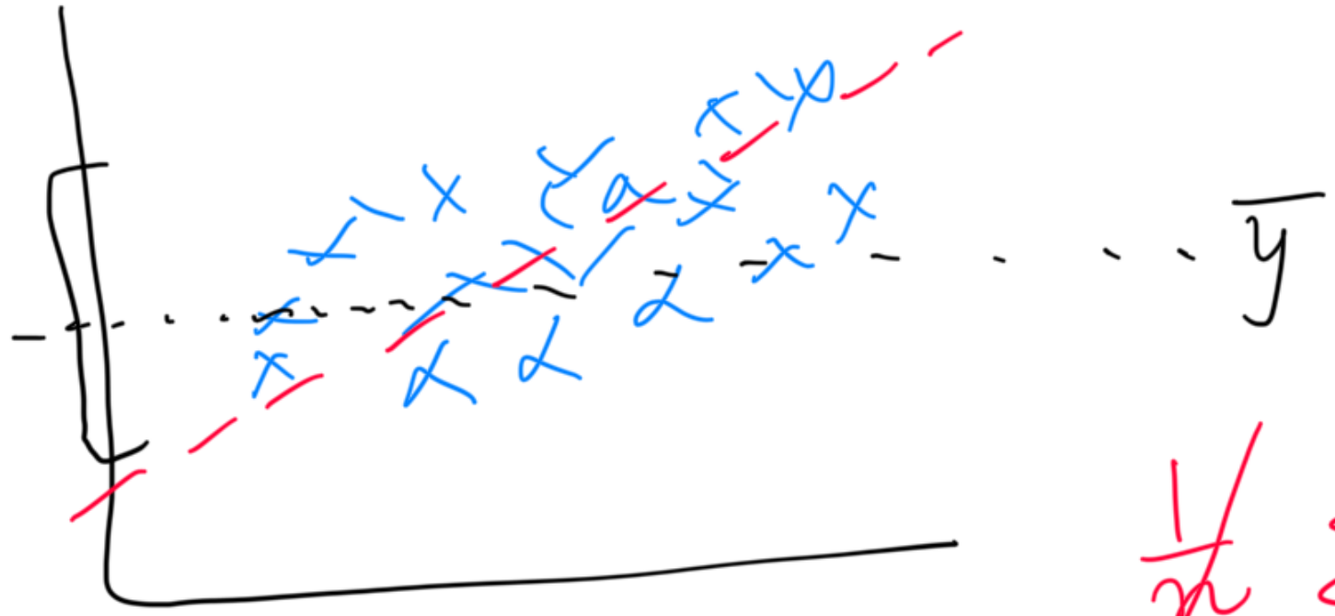
good ??

MSE_R

MSE_T

org.

SSE



$$\cancel{\frac{1}{n}} \sum (y - \bar{y})^2$$

$$\frac{1}{n} \left[(y - \hat{y})^2 \right]$$

$$\left[\frac{SSE_T - \overset{\to 0}{SSE_R}}{SSE_T} \right] \rightarrow \boxed{\text{r-squared value}}$$

$$R^2 = \boxed{1 - \frac{SSE_R}{SSE_T}}$$

goodness of fit

good $\approx$ 0.8 $\approx$

$R^2 = (-\infty, 1]$

< 0.5

bad model